Band-1 Hadamard Order 2404 — Miyamoto (H_600 Paley-I) + Paley-II Cross-Check; Gap 2052 Still OPEN

H_600/C_602/H_2404 PASS (Miyamoto + Paley-II); 2404 in SPK via Paley II q=1201; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_2308_Miyamoto`; `2026-09-14_Band1_Hadamard_2276_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_2404_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 600 and 2404 only** (verified max $|HH^\top - nI| = 0$), via **two independent paths** for order 2404 (Miyamoto primary; Paley type II cross-check). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 2404 **enters SPK** via Paley II ($q = 1201$); it is **not** a Miyamoto-only bonus (contrast prior 2276/2228/2164). Honest SPK audit: Paley-I **FAIL** ($q = 2403$ composite); Kronecker **FAIL** ($2404 = 4 \cdot 601$, 601 not Hadamard-admissible); Paley-II **PASS**. `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 2308 on box. This increment constructs the **next preferred** Miyamoto-eligible target after $q = 577$: **order 2404** ($2404 = 4 \cdot 601$, 601 prime $\equiv 1 \pmod 4$). Seed H_{600} via Paley type I ($q = 599$ prime $\equiv 3 \pmod 4$). Symmetric Paley conference C_{602} over prime 601. $H_{2404} = \text{Miyamoto}(H_{600}, C_{602})$. Independently, H_{2404} also verifies via **Paley type II** with $q = 1201$ (explains SPK membership). Unit tests **23/23 PASS**. SPK next gap remains **2052** (OPEN). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 2404; Miyamoto; Paley type I; Paley type II; Paley conference; constructive gap 2052; Band-1; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in $\{\pm 1\}$ satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{600} constructed and verified (Paley type I, $q = 599$). 2. C_{602} constructed and verified (Paley conference, $q = 601$): $CC^\top = 601I$, zero diagonal. 3. H_{2404} constructed and verified (Miyamoto). **Also** constructed and verified via Paley type II ($q = 1201$). **Not** the Hadamard conjecture. 4. SPK: $2404 \in$ SPK (via Paley II). Paley-I and Kronecker paths for 2404 **rejected**. Next constructive gap remains **2052** (OPEN).

2. Arithmetic and eligibility

Quantity	Value
Target order	$2404 = 4 \cdot 601$
$q = 601$	prime, $601 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{600}
Needed conference	order $602 = q + 1$
H_{600} path (used)	Paley type I, $q = 599$ (prime $\equiv 3 \pmod 4$)
Conference path	Symmetric Paley over prime 601
SPK path for 2404	Paley type II, $q = 1201$ prime $\equiv 1 \pmod 4$; order $2(q + 1) = 2404$
Paley-I for 2404	**REJECTED**: $q' = 2403$ not prime
Kronecker for 2404	**REJECTED**: only 4×601; $601 \not\equiv 0 \pmod 4$ (not admissible)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{600}	Paley type I, $q = 599$	0	0.2930	**PASS**
C_{602}	Paley conference, $q = 601$	0	0.2684	**PASS**
H_{2404}	Miyamoto(H_{600}, C_{602})	0	6.1734	**PASS**
H_{2404} P-II	Paley type II, $q = 1201$	0	6.8486	**PASS**

Primary Miyamoto pipeline wall (incl. unit tests + SPK refresh + Paley-II cross-check) $\approx$ 14.37 s. Compressed ± 1 matrices saved as `H600_pm1.npz`, `C602_pm1.npz`, `H2404_pm1.npz`, `H2404_paleyII_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible $\leq 2000 \cup$ on-box prior ships $\{508, 528, 540, 556, 568, 284, 24, 576, 2004, 2036, 2116, 2164, 2228, 227$ $\cup$ verified $H_{600} \cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4000). **Miyamoto is not folded into SPK**; order 2404 enters SPK **because** Paley II applies ($q = 1201$).

Quantity	Value
Known count	880
2404 in SPK	**True** (Paley II $q = 1201$)
600 in SPK	True (Paley I $q = 599$)
2308 in SPK	True (prior; Paley II $q = 1153$)
2276 in SPK	False (prior Miyamoto bonus)
Next standard gap	**2052** (OPEN)
Missing first (sample)	2052, 2060, 2076, 2092, 2108, 2148, 2156, 2172, 2212, 2284...

5. Gap 2052 (still OPEN; non-finding ONLY)

This ship does **not** fill gap 2052. Prior triage stands: Paley-I/II / Kronecker / Miyamoto-Paley-conf all **FAIL** for order 2052. Claim level for 2052 remains method triage / non-finding ONLY. Non-finding $\neq$ counterexample; Hadamard conjecture remains OPEN. No H_{2052} matrix is claimed.

6. Unit tests (23/23)

Test	Result
`H20_paley_I_control`	PASS
`H4_sylvester`	PASS
`paley_II_q5_H12` / `paley_II_q13_H28`	PASS

Test	Result
`miyamoto_H20_from_H4_conf6`	PASS
`paley_I_q11_H12`	PASS
`q601` / `n2404` / `q599` / `H600` / `C602` formulae	PASS
`q1201` / Paley-II order-2404 formula	PASS
`paley_I_H2404_path_rejected`	PASS
`kronecker_H2404_path_rejected`	PASS
`reject_all_ones_H8`	PASS
`conference_6_paley_control`	PASS
`miyamoto_eligible_2404_arith`	PASS
`conference_602_shape_expected`	PASS
`gap_2052_not_filled_by_this_ship`	PASS

7. Next optional targets

Miyamoto-eligible primes $q > 601$, $q \equiv 1 \pmod 4$ (sample):

q	$n = 4q$	Need H_{q-1}	On-box path notes
613	2452	612	no quick SPK seed noted
617	2468	616	Double(Paley I half 308)
641	2564	640	Kronecker(4×160)
661	2644	660	**Paley I** ($q = 659$)
673	2692	672	Kronecker(4×168)

Preferred next. Method-attack remaining SPK gap **2052** (GS/SDS for order ~513 if literature seeds appear), **or** construct H_{2468} / H_{2644} via listed seeds, **or** Track B / Object 1 await. GATE-04 remains LOCKED.

8. Non-claims (explicit)

- Full Hadamard conjecture: **OPEN**
- Clay-closed / finished-ToE: **false / not claimed**
- Gap 2052 filled: **NOT claimed** (OPEN)
- GATE-04: **LOCKED**
- RH / P≠NP / YM / Frankl / NS-solved / AC-solved: **not claimed**
- Miyamoto-only bonus for 2404: **false** (in SPK via Paley II)

9. Reproducibility

```
cd /workspace/experiments/band1/hadamard_2404_miyamoto
python3 verify_hadamard_2404.py
# expects: Unit tests 23/23; H600/C602/H2404 Miyamoto PASS; Paley-II cross-check PASS
# artifacts: results.json, literature_ledger.json, SUMMARY.txt, *.npz
```

Lab path. /workspace/experiments/band1/hadamard_2404_miyamoto/
PDF shelf. /workspace/DOWNLOAD_PDFs/2026-09-15_Band1_Hadamard_2404_Miyamoto.pdf